SIEVESPACE V2

SIEVE

GitHub, redesigned for maintainers.

> "Sieve helps maintainers understand pull requests, identify risks, and decide what needs attention."

- Standard-Library Python Engine
- ⚡ Gemma 4 Visual Intelligence
- 🔒 100% Read-Only on GitHub
- 🛡️ Contributor Code Never Runs
- 📦 MIT Open Source

Open source is drowning in pull requests that look convincing.

Maintainers spend evenings reviewing code changes without knowing what they really touch.

Volume & Low-Effort PRs

Repositories receive dozens of pull requests. A recent Linux Foundation survey found 25% of incoming pull requests are AI-generated slop that looks polished but does nothing.

Reviewing Code Takes Hours

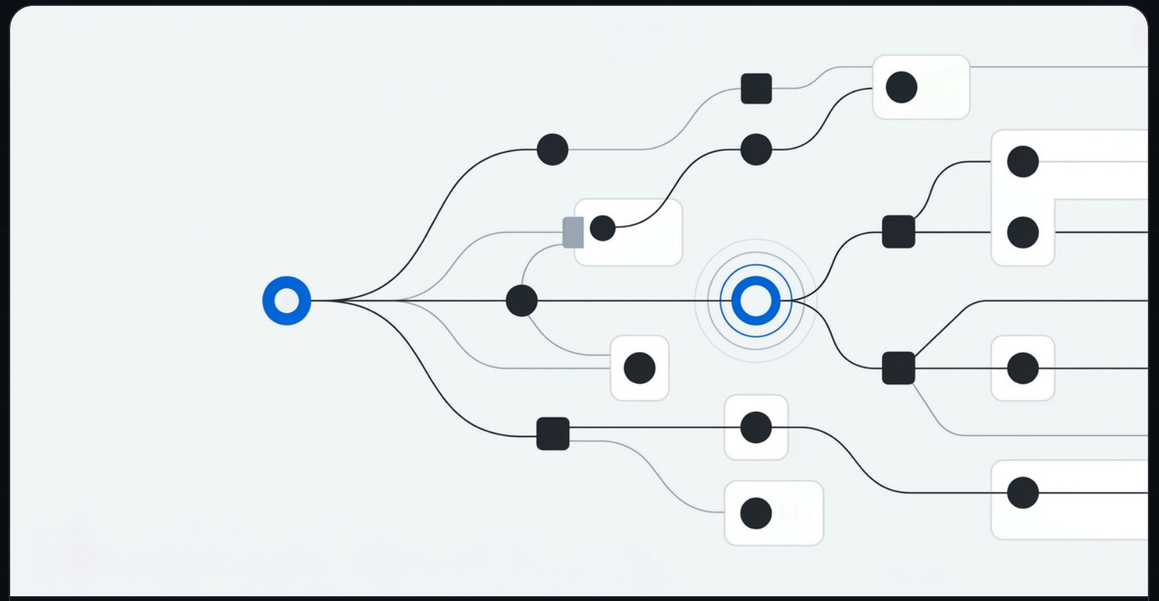
Maintainers manually trace diffs, check linked issues, and verify tests. Reading through superficial changes burns valuable volunteer maintainer time.

Hidden Ripple Effects

A change in one file can quietly break downstream modules three directories away. Standard GitHub diffs only show modified lines, not the affected repository topology.

Maintainers Need to Know What to Review First

GitHub lists pull requests by timestamp, not risk or readiness. Maintainers have no fast way to identify which pull requests are safe and which are dangerous.



The Blast Radius Blindspot: GitHub shows you what text changed. Sieve maps the entire import graph so you immediately understand what that change could break.

From incoming pull request to an evidence-backed maintainer brief.

A clean 5-step workflow that turns raw code diffs into clear, prioritized maintainer decisions.

STEP 01

Connect

Chrome extension attaches to GitHub pull request pages. 100% read-only; never requests write tokens or posts comments automatically.

STEP 02

Understand

Standard-library Python engine statically parses the repository AST into an internal import graph (Python & JS/TS).

STEP 03

Analyse

Verifies contributor claims: does the linked issue exist? Are claimed tests in the diff? Do new imported modules actually exist?

STEP 04

Identify Risks

Computes blast radius DAG: modified files, direct dependents, test consumers, and regression radar for removed or renamed symbols.

STEP 05

Guide Action

Surfaces actionable queues (Action Needed vs Blocked), generates exact pytest reproduction commands, and drafts polite reviewer feedback.

VERIFIED CODE FACTS **Deterministic AST Analysis**

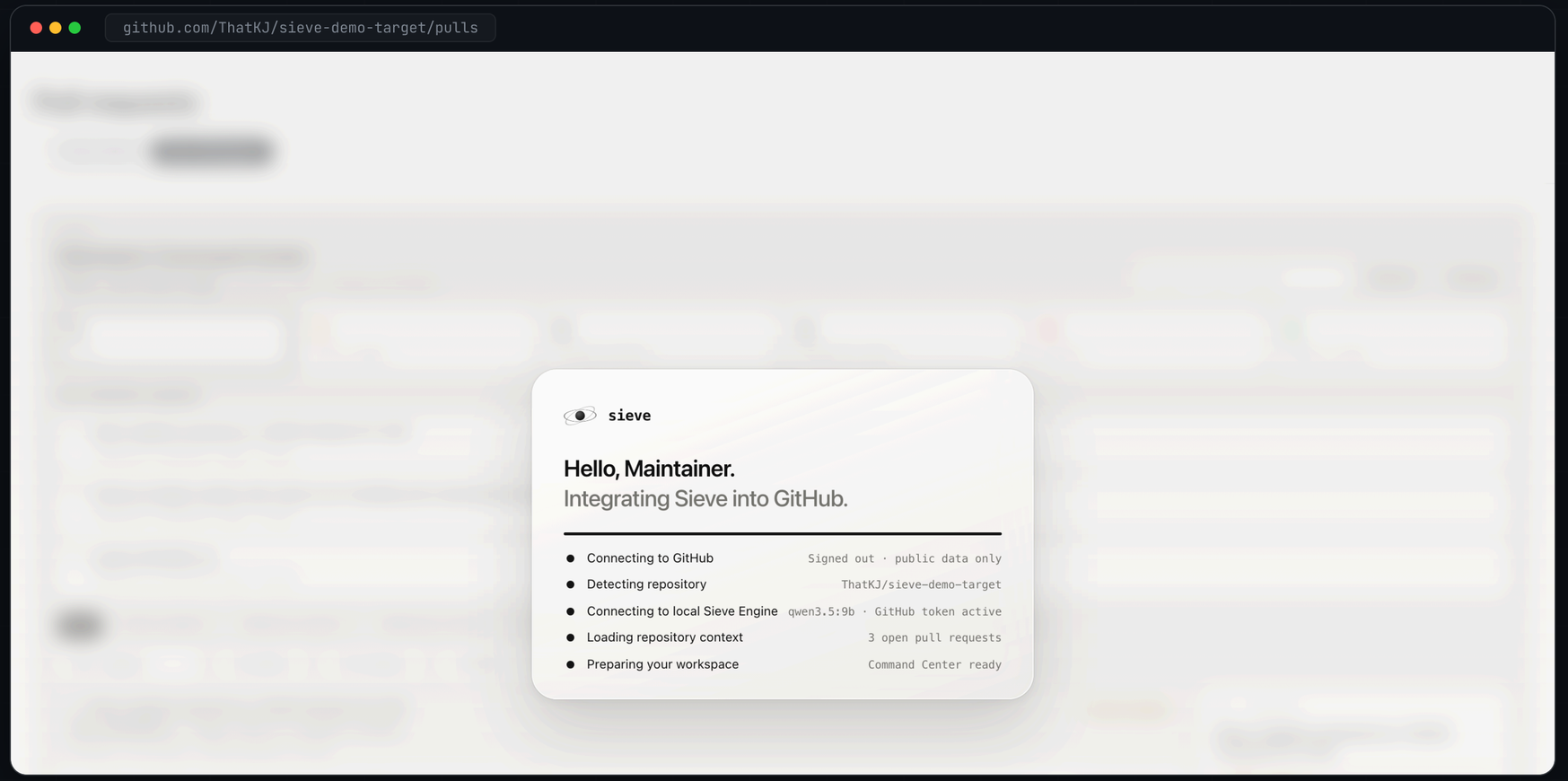
Standard-library Python AST parsing. Evaluates real file existence, exact import lines, deleted symbol references, and diff stats. 100% deterministic, 0% hallucinations. Never executes untrusted contributor code.

ARCHITECTURAL REASONING **Gemma 4 Visual Intelligence**

Google Gemma 4 multimodal model (`gemma-4-26b-a4b-it`) receives the rendered graph PNG and structured AST evidence. Explains architectural implications and missing tests. Strictly badged as model hypothesis.

GitHub becomes a maintainer workspace.

Live Chrome extension injecting a high-signal command center directly into GitHub's pull requests page.



NON-DESTRUCTIVE TOGGLE

Native GitHub Switcher

Injected into GitHub's subnav bar. Maintainers toggle instantly between native GitHub and Sieve Workspace with zero page reloads.

TURN-TAKING LANES

Triage KPI Strip

Automatically sorts PRs by responsibility: *Action Needed: You*, *Waiting on Author*, *Blocked*, and *Merge Ready*.

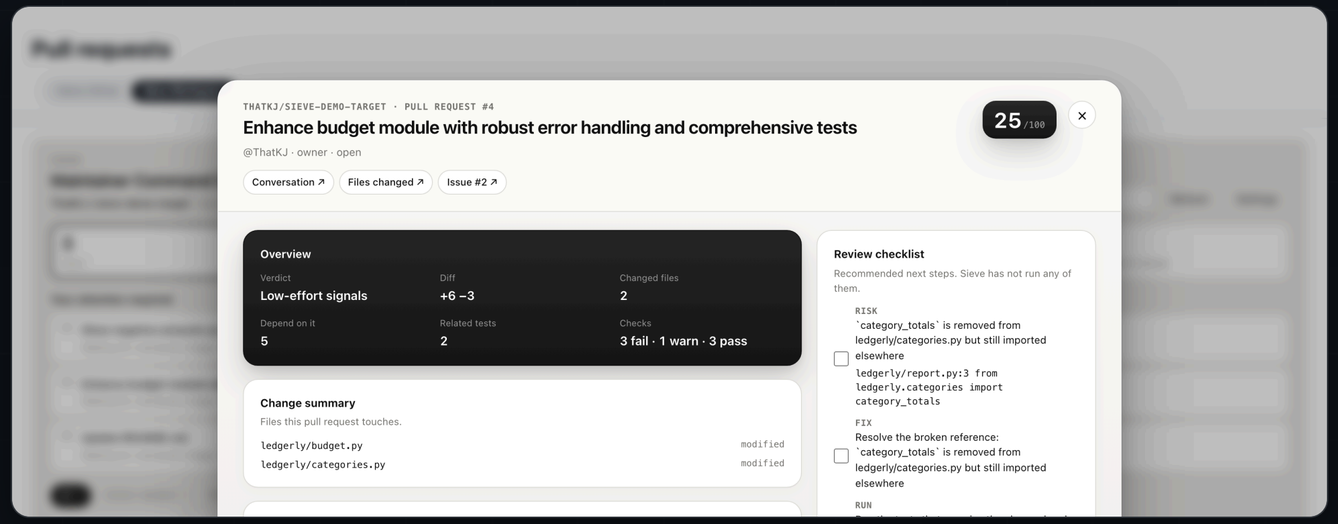
RISK PRIORITIZATION

AST Blast Radius Ranks

Pull requests are ranked by real architectural impact score rather than submission timestamps. High-risk changes surface immediately.

Sieve doesn't just show what changed. It explains what it affects.

Deep architectural inspection of ThatKJ/sieve-demo-target#4 (claims tests & bugfix, but introduces breakages).



SCORE: 25 / 100 · LOW-EFFORT

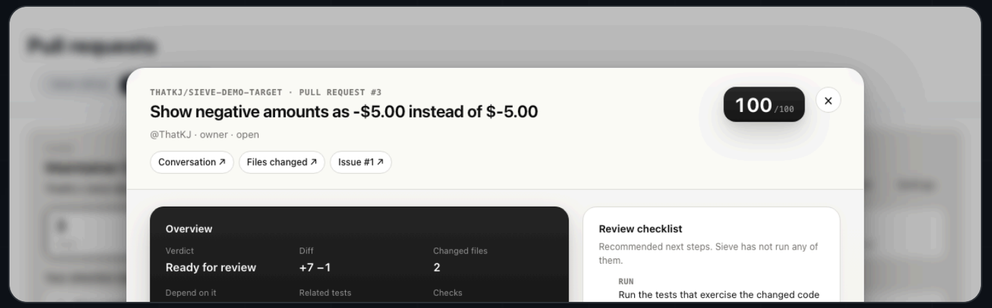
PR #4 Analysis

FAIL **Claimed Tests Missing:** Description says “Added comprehensive unit tests”, but the diff contains 0 test files.

FAIL **Phantom Imports:** Code imports `ledgerly.validators`, which does not exist anywhere in the repository.

RADAR **Regression Alert:** Renamed `category_totals`, but `report.py:3` still imports it. Downstream breakage flagged!

AST **Blast Radius DAG:** 2 modified files, 5 direct dependent modules, 2 related test files identified.



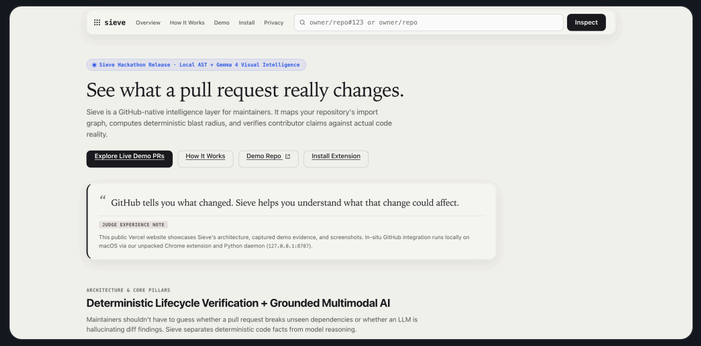
The Maintainer Result: An actionable verification brief, exact pytest reproduction commands, and a polite contributor checklist. Nothing is posted; Sieve is 100% read-only.

Understand every change. Review with confidence.

A working, local-first open-source intelligence layer for GitHub maintainers.

Web Showcase

Deployed



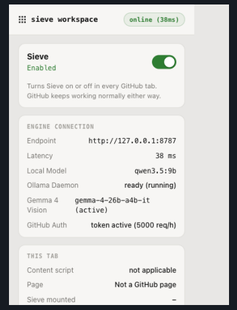
Public architecture showcase with interactive 3D Repository Atlas, demo PR explorer, and full documentation.

`URL: sieve.vercel.app / local`

`Atlas: 3D force-directed import DAG`

Chrome Extension

Manifest V3



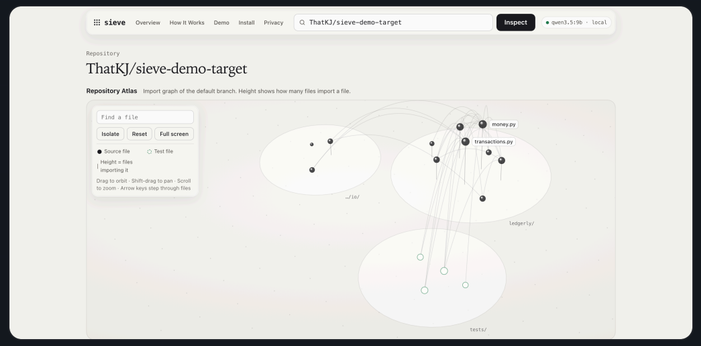
Connects to local daemon in 4ms. Verifies Ollama (qwen3.5:9b), Gemma 4 multimodal model, and GitHub API quota.

`Status: Online (4ms latency)`

`Model: gemma-4-26b-a4b-it active`

Local Engine & Open Source

MIT License



Zero third-party dependencies required. Pure standard-library Python daemon. 94 automated unit & lifecycle tests.

`Repo: github.com/ThatKJ/sieve`

`Demo: github.com/ThatKJ/sieve-demo-target`

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- ✓ Zero Cloud Lock-In
- ✓ Untrusted Code Never Runs
- ✓ Open-Weight AI